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CSA0711

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I Dinesh Bhavadwaj (192572028) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
 I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Bhavadwaj

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	2	2	2	2
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1
		8	8	8	6	6

Total Marks 36 /40

36/40
HeP

[Signature]
 Signature of the Faculty

Co2-AT-1

1. Subnetting for University Network (192.168.20.0/24)

Given:

- Network: 192.168.20.0/24
- Lab A → /26
- Lab B → /27
- Lab C → /28

Sol)

<u>Lab</u>	<u>Subnet Mask</u>	<u>Network Address</u>	<u>Valid Host Range</u>
Lab A	/26 (255.255.255.192)	192.168.20.0	192.168.20.1 192.168.20.62
Lab B	/27 (255.255.255.224)	192.168.20.96	192.168.20.97 192.168.20.110
Lab C	/28 (255.255.255.240)	192.168.20.96	192.168.20.97 192.168.20.110

<u>Broadcast Address</u>	<u>Usable Hosts</u>
192.168.20.63	62
192.168.20.95	30
192.168.20.111	14

Host calculation

- $126: 2^6 - 2 = 64 - 2 = 62$
- $127: 2^5 - 2 = 32 - 2 = 30$
- $128: 2^4 - 2 = 16 - 2 = 14$

Final Answer

- Lab A: Network = 192.168.20.0
Broadcast = 192.168.20.63, Hosts = 62
- Lab B: Network = 192.168.20.64
Broadcast = 192.168.20.95, Hosts = 30
- Lab C: Network = 192.168.20.96
Broadcast = 192.168.20.111, Hosts = 14

Subnet Analysis

Given

- System 1: 192.168.1.130/26
- System 2: 192.168.1.190/26
- Subnet Mask = 255.255.255.192

A /26 Subnet increments by 64:

- 0-63
- 64-127
- 128-191
- 192-255

System 1

- IP: 192.168.1.130
- Network Address: 192.168.1.128
- Broadcast Address: 192.168.1.191